Topic Best Practices: Final

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Document Revisions Table

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1.0	Initial Document	Professor Clark/Professor Porter/Professor Steiner	6/5/2026
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Topic

Topic Name
A Study on the Influence of Thematic Affordances on Players' Cognitive Map Construction

Hypothesis
This study hypothesizes that thematic affordances significantly influence how players understand and construct cognitive representations of interior environments.

Specifically, players first identify the function of space through environmental objects, architectural features, visual themes, and spatial design cues. Rather than relying only on geometric layouts or route memorization, players use functional understanding of spaces as cognitive anchors to organize spatial information.

Engine Preference(s)
Starfield Creation Kit

Topic Description
Thematic affordances refer to the ability of an environment to communicate functional meaning through architectural design, visual themes, environmental storytelling, object placement, and spatial composition. Rather than relying on explicit instructions or interface elements, players infer the purpose and expected use of space directly from the environment itself.

In this study, a large-scale interior level is treated as a complex interior environment composed of multiple interconnected rooms. Each area possesses a distinct functional identity and thematic purpose, such as commercial area, dining area, entertainment area and so on.

These spaces are connected through corridors, staircases, elevators, and a central atrium, forming a unified spatial system.

This study focuses on how thematic affordances influence players' spatial understanding through two sequential layers of cognition.

1. Functional Readability

This layer examines whether players can identify the function of a space through thematic environmental cues.

Clear thematic affordances enable players to accurately identify the function and purpose of a space.

Through environmental elements such as architecture, objects, visual themes, and environmental storytelling, players can infer whether a space serves functions such as engineering, dining, commerce, accommodation, or transportation.

This functional understanding provides meaningful labels for individual spaces and strengthens players' memory of explored locations.

Examples include:

Engineering Area

- Pipe systems
- Control terminals
- Maintenance equipment

Dining Area

- Dining tables
- Kitchen facilities
- Food displays

The primary objective is to determine whether players can correctly understand:

- What type of space they are in
- What the primary function of that space is

and whether this functional understanding remains stable throughout subsequent exploration.

2. Cognitive Mapping

This layer examines whether players can use functional zones to construct an overall understanding of the environment.

After identifying the functions of individual spaces, players use these functional identities as spatial reference points to construct a broader cognitive map of the environment.

Through spatial identification and spatial revisitation, players gradually establish positional relationships between functional areas and integrate local spatial knowledge into a coherent understanding of the overall environment.

The focus is not on route memorization, but rather on whether players can understand:

- The positional relationships between functional areas
- The connections between functional regions
- The organizational structure of the overall environment

For example, whether players understand that:

- The Shopping Mall is located near the Dining Area

- The Industrial Area is located on lower levels
- The Commercial Area serves as a hub connecting multiple regions

Through exploration of multiple miniscapes, players gradually integrate local functional knowledge into a comprehensive understanding of the entire spatial system.

Research Goal

This study aims to investigate how thematic affordances influence players' understanding of complex interior environments and to establish a cognitive framework that explains the progression from functional recognition to global spatial understanding.

Specifically, this study explores:

- How players identify the function of space through environmental cues
- How players use functional spaces for spatial localization and construct a global cognitive map of the environment

Expected Contribution

This study aims to establish a thematic-affordance-based framework for spatial cognition and explore:

- How thematic affordances support functional space recognition.
- How functional spaces serve as spatial anchors for player localization and cognitive map construction.

The findings are expected to provide design insights into a variety of complex interior level types, such as cruise ship environments, space stations, and large-scale indoor exploration spaces.

Topic Best Practices & Research Questions

Researched Best Practices

1. By placing consistent thematic objects, architectural elements, and visual design features within the same functional space, designers can establish a clear thematic identity for that area.
    a. Yang, Anona "Best Practices: Reusing Space to Enhance Narrative Immersion"
2. The repeated environmental cues help players infer the intended function of the space and reinforce functional recognition.
3. By revisiting spaces and identifying their functions through "thematic affordances," players can reinforce their spatial orientation and understanding of the overall environment.

Relevant Research Questions

1. How do consistent thematic objects, architectural elements, and visual design features influence players' ability to identify the function of space?
2. How do repeated thematic affordances affect players' recognition and memory of functional spaces?
3. How do players use functional identities established through thematic affordances as cognitive anchors to construct an overall understanding of a level's spatial structure?

Research Source(s)

[1] NIC PHAN|GAME DESIGNE, “Affordances in Game Level Design”, Affordances in Game Level Design

Description

Game design uses cultural and inferred affordances to guide players’ perception of interaction possibilities.

[2] Yang, Anona “Best Practices: Reusing Space to Enhance Narrative Immersion”

Description

Talks about mental mapping stuff, which is basically thematic affordance.

[3] Rodriguez, Ian “Predicting Player Perception of a Room’s Purpose through Context Complexity”

Description

Investigates how much contextual environmental information is required for players to correctly infer a room’s intended function in a level design context.

[4] Medium, “Mastering the Invisible: How Affordances & Signifiers Shape Player Experience in Level Design”, Mastering the Invisible : How Affordances & Signifiers Shape Player Experience in Level Design | by Genesis | Medium

Description

Affordances define what can be done in a game environment, and signifiers indicate how and where to do it.